

The Read-Budget of an L-Function

$\sqrt{\cdot}$ (Analytic Conductor) as the Universal Cost of a Read, from Elliptic Curves to the Critical Line — with the Rank, the Coefficient, and the Conductor Sorted into Their Layers

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Status. Every quantitative statement was regenerated from local data (point counts a_p , or the Riemann–Siegel main sum) within the working session. Labels: LOCKED (a classical theorem, or a measured fact already independently reproduced), PULL (measured here on one machine, awaiting independent re-run), Ω (coherent but with no contact point to measurement). This paper does not claim to prove any Millennium problem. Its results are a clean location — of where the conductor of an L-function lives in a computation, and of a cost law that is the same across two independent L-function families — together with a disciplined separation of what the accompanying interpretive framing does and does not establish.

1. Abstract

Reading an L-function is not free, and the cost is structured. This paper measures the cost and locates its source. An elliptic curve is read through two channels on its local data a_p : an order channel (the Mestre–Nagao weighting) that returns the analytic rank as a monotone bias, and a coefficient channel (the approximate functional equation) that returns the leading Taylor coefficient $L^{(r)}(E, 1)/r!$. Both are taken here to rank four, with the coefficient matching LMFDB to 10^{-10} for ranks zero through three and to six digits at rank four. The cost of these reads is set by a third quantity, the conductor, and it obeys a clean law: the read-budget — the number of terms that must be summed before the answer stabilizes — scales as the square root of the conductor. Across four orders of magnitude in conductor the ratio of terms to $\sqrt{N}$ is flat at about six, with a fitted exponent of 0.505.

That law is not special to elliptic curves. The same measurement on the Riemann zeta function, through the Riemann–Siegel main sum, gives a read-length of $\sqrt{t/2\pi}$ — the square root of zeta’s analytic conductor at height t — with the main-sum error tracking $t^{-1/4}$ and the fitted exponent approaching one half. The budget is therefore the universal analytic-conductor principle: the cost of reading any L-function is the square root of its analytic conductor, a property of the L-function itself and not of the algorithm running on it. This is the

rigorous form of an intuition that recurred in the surrounding discussion — that the conductor is the size of the memory a curve needs to compile its rank — and it forces a correction to a companion proposal. The cost does not live in the sub-floor exhaust of the factorization: that exhaust was measured, in prior work, to be a function of the algorithm and precision that decorrelates across algorithms and is blind to the object's value, so it cannot carry a conductor. The friction is the $\sqrt{\text{conductor}}$ read-length, an object quantity; the exhaust fingerprints only the machine. The paper closes by sorting the three arithmetic quantities into their layers — rank and coefficient as object channels, conductor as the budget, sub-floor as machine exhaust — and by separating, explicitly, the measured core of the “instruction-set” reading of these results from its interpretive extension.

2. The question: what does it cost to read an L-function, and where is the conductor?

An L-function is defined by an infinite series, and no infinite series is summed in practice. What is summed is a finite piece, made to converge rapidly by folding the series against its functional equation — the approximate functional equation for an arithmetic L-function, the Riemann–Siegel formula for zeta. The length of that finite piece is the read-budget, and it is not a free parameter. It is fixed by the analytic conductor of the L-function, and the purpose of this paper is to measure that dependence cleanly, in two independent families, and then to say precisely what follows.

The motivating context is a recurring proposal in the surrounding work: that the conductor N of an elliptic curve is a kind of friction or cost, and that this cost should be visible somewhere in the computation. That intuition is correct, and one companion claim about *where* the cost lives is not. The claim was that the friction should appear in the sub-floor exhaust of the Cholesky factorization — the stream of sign bits below the precision floor, the “debris field” of the computation hitting its wall. Prior measurement in this program settled the character of that exhaust: the sub-floor sign stream is a deterministic function of the algorithm and the precision jointly, it decorrelates to chance under a change of algorithm and under a one-digit change of precision, and it is blind to the object's own value. It is a hash of the machine configuration, not a property of the object. An object property like the conductor therefore cannot be read from it. This paper locates the conductor where it actually is — in the read-budget — and the location is a clean square-root law.

3. The two channels of BSD, read to rank four

Before the cost, the two things whose cost is being paid. An elliptic curve E over the rationals presents its arithmetic through the sequence $a_p = p + 1 - \#E(\mathbb{F}_p)$, and the Birch–Swinnerton-Dyer conjecture ties two invariants of E to the behaviour of $L(E, s)$ at the central point $s = 1$: the analytic rank (the order of vanishing) and the leading Taylor coefficient. These are read through two different slots on the same a_p .

The **order channel** is the Mestre–Nagao statistic $S(X) = -\frac{1}{\log X} \sum_{p \leq X} a_p \frac{\log p}{p}$, the explicit-formula shadow of the order of vanishing, which returns the rank as a monotone bias. Across five curves of ranks zero through four — 11a1, 37a1, 389a1, 5077a1, and 234446a1 — the ladder at $X = 20000$ is $-0.22, 0.65, 1.83, 2.92, 3.34$, monotone through rank four, with rank four the *loudest* read rather than the

faintest. [PULL, verified detector] The Weil bound $|a_p| \leq 2\sqrt{p}$ — the Riemann Hypothesis over each $\mathbb{F}_p$ — holds at every good prime for all five curves, with $\max |x_p| \rightarrow 0.997$, confirming the shape channel these traces feed. [LOCKED, classical]

The **coefficient channel** is the approximate functional equation

$$L^{(r)}(E, 1)/r! = 2 \sum_{n \geq 1} \frac{a_n}{n} G_r\left(\frac{2\pi n}{\sqrt{N}}\right), \quad G_r(x) = \frac{1}{(r-1)!} \partial_s^{r-1} [x^{-s} \Gamma(s, x)]|_{s=0},$$

with $G_0(x) = e^{-x}$ and $G_1(x) = E_1(x)$. The incomplete-gamma form of G_r replaces the slow numerical integration of earlier implementations and matches it to 10^{-31} ; the a_n are built from the a_p by the Hecke recurrence. Summed with adaptive truncation, the leading coefficient reproduces the LMFDB values: 0.2538418609 (rank 0), 0.3059997738 (rank 1), 0.7593165 (rank 2), 1.7318499 (rank 3) to relative error $\sim 10^{-10}$, and 8.9438474 (rank 4) to 5.3×10^{-6} at reduced precision. [PULL; LMFDB is the external anchor] Both invariant channels therefore read to rank four. What makes rank four costly is neither the order (the loudest read) nor the coefficient value (computed here) but the number of terms the coefficient sum required — 2144 for the rank-four curve — and that number is the subject of the next section.

4. The read-budget: $\sqrt{N}$ for elliptic curves

The read-budget is the number of terms of the coefficient sum needed before the tail falls below a fixed floor. It is a property of the curve, and it scales cleanly with the conductor. Measuring the length of the sum $\sum (a_n/n) e^{-2\pi n/\sqrt{N}}$ needed for the tail to drop below 10^{-1} , across the five curves:

curve	conductor N	read length (terms)	terms $/\sqrt{N}$
11a1	11	20	6.03
37a1	37	36	5.92
389a1	389	122	6.19
5077a1	5077	450	6.32
234446a1	234446	3003	6.20

The ratio terms $/\sqrt{N}$ is flat at about six across four orders of magnitude in N , and a fit of terms $= c N^\alpha$ returns $\alpha = 0.5046$ against the prediction 0.5. [PULL] The read-budget is $O(\sqrt{N})$: the conductor is the size of the memory the curve needs to compile its answer, and the compilation length is the square root of the conductor. The exponential decay of the kernel is set by the scale $2\pi/\sqrt{N}$, so the number of terms before the kernel drops below the floor is proportional to $\sqrt{N}$ times the number of digits — which is exactly the flat ratio observed. This is the mechanism behind the “cache size” reading of the conductor: not a metaphor but a measured length, and the length is $\sqrt{N}$.

The rank-four gate that has stood open in this program — the certification of $L^{(4)}(E, 1)/4!$ under the Stieltjes instrument, named E-P6 for its compact-support window — is now located precisely. It is not that rank four is harder *arithmetic*, and not that the coefficient value is inaccessible; it is that the conductor $N =$

234446 demands a read-length of thousands of terms, and that length is what meets the instrument’s compact-support window. The gate is a budget constraint, and naming it as one is what makes it tractable to reason about: the question is not “why is rank four hard” but “how does the instrument accommodate a $\sqrt{N}$ -length read when N is large.”

5. The same law for zeta: Riemann–Siegel and $\sqrt{t/2\pi}$

If the read-budget were a property of the elliptic-curve algorithm, it would have no reason to appear elsewhere. It appears identically for the Riemann zeta function, which is the cleanest possible cross-check because zeta sits at the centre of the other Clay problem this program orbits.

Zeta on the critical line is read through the Riemann–Siegel formula, whose main sum is

$$Z(t) \approx 2 \sum_{n=1}^N \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}}, \quad N = \lfloor \sqrt{t/2\pi} \rfloor,$$

where θ is the Riemann–Siegel theta function and $Z(t)$ is the Hardy function, $|Z(t)| = |\zeta(1/2 + it)|$. The main sum approximates $Z(t)$ to an accuracy of order $t^{-1/4}$, and its length is $\sqrt{t/2\pi}$ — the square root of zeta’s analytic conductor at height t , which is $t/2\pi$. Measured:

height t	N terms	main-sum error	$t^{-1/4}$	$N/\sqrt{t/2\pi}$
100	3	0.453	0.316	0.752
1000	12	0.114	0.178	0.951
10000	39	0.111	0.100	0.978
30000	69	0.0858	0.0760	0.999
100000	126	0.0549	0.0562	0.999

The error tracks $t^{-1/4}$ across three orders of magnitude, the ratio $N/\sqrt{t/2\pi}$ converges to one, and a fit of $N = c t^\alpha$ returns $\alpha = 0.536$, approaching one half as t grows (the excess is the finite- t contribution of the low points). [LOCKED — Riemann–Siegel is classical; the length and accuracy are measured] Zeta’s read-length is $\sqrt{t/2\pi}$, the elliptic curve’s is $\sqrt{N}$, and both are the square root of the analytic conductor.

6. The universal analytic-conductor principle

Two families, one law. The read-budget of an L-function is the square root of its analytic conductor: $\sqrt{N}$ for the L-function of an elliptic curve at the central point, $\sqrt{t/2\pi}$ for zeta at height t . This is not a discovery of this paper — it is the analytic-conductor principle of analytic number theory, the statement that the analytic conductor governs the length of the approximate functional equation for every L-function in the family. What this paper contributes is a clean measurement of it across two independent families, and the use of that measurement to settle where the conductor lives in a computation.

The consequence is exactly the correction the surrounding discussion needed. The read-budget is *object-determined*: it scales with the conductor, a property of the L-function, and it does not depend on the precision or the algorithm — the ratio terms/ $\sqrt{N}$ is flat regardless of the working precision, and the same $\sqrt{\text{conductor}}$ law holds for two families computed by entirely different methods (a Hecke-recurrence Dirichlet sum for the curve, a cosine main sum for zeta). The sub-floor exhaust, by contrast, is *machine-determined*: it decorrelates across algorithms and precisions and carries none of the object’s value. The conductor is in the budget, not the exhaust, because the budget is opcode geometry — intrinsic to the L-function — and the exhaust is assembly debris — intrinsic to the machine. The friction the discussion was chasing is real, and it is fetch length, not execution noise.

7. The three-layer sort on arithmetic

The whole program has been a sorting of quantities into the layer they belong to, and the BSD side now sorts cleanly:

quantity	channel	layer	how it is read
analytic rank	order (Nagao)	object	monotone bias, loudest at rank 4
leading coefficient	coefficient (AFE)	object	approximate functional equation, to rank 4
conductor	budget	object (cost)	$\sqrt{N}$ read-length, $\alpha = 0.505$
computed value	assembly	frame	the number a particular precision returns
sub-floor signs	exhaust	machine	hash of (algorithm, precision), object-blind

The rank and the coefficient are the two BSD invariants, read through two channels. The conductor is not a third invariant read but the *budget* that both reads cost — an object quantity, since it is fixed by the L-function, but a cost rather than a value. Below the object layer sit the frame (the specific computed value at a given precision) and the exhaust (the sub-floor fingerprint). The recurring proposal to read the conductor from the exhaust fails because it crosses layers: it looks for an object cost in the machine debris. The conductor is one layer up, in the read-budget, and the $\sqrt{N}$ law is where it is written.

8. The instruction-set reading: measured core and interpretive extension

The surrounding discussion has developed an elaborate reading of these results as an instruction-set architecture — the L-function as an opcode sequence, the rank as an instruction pointer, the coefficient as a register, the conductor as a cache size, the sub-floor as a NOP trace, the wall as a HALT boundary — and it is worth being exact about which parts of that are measured and which are interpretation, because the distinction is the difference between a defensible result and an overreach.

The **measured core** is the three-layer sort of Section 7, and it is real and reproducible. There is an object layer whose quantities are invariant under the machine that reads them; there is a frame layer of specific computed values; and there is an exhaust layer that fingerprints the machine and is blind to the object. The

conductor sits in the object layer as a cost; the sub-floor sits in the exhaust layer as a machine hash. Relabelling these layers with the vocabulary of a processor — opcode, register, cache, NOP — is harmless and sometimes clarifying, and the specific mappings above are consistent with the measurements. The claim that the conductor is a “cache size” is, made precise, the $\sqrt{N}$ budget law, which is measured.

The **interpretive extension** is the claim that this instruction-set architecture *is* the shape of the measurements themselves — that the universe is literally a distributed compiler, that its laws are an opcode, that matter is saturated execution, and that the halting problem is thereby dissolved. None of that follows from the measurements, and it is marked as interpretation, Ω . The three-layer sort is a fact about how these particular arithmetic objects are read; extending it to an ontology of the physical universe is a reading of that fact, not a consequence of it. The halting problem in its technical sense is a specific undecidability result about Turing machines, and “run is the middle” — however apt as a description of a computation that neither completes nor faults — does not resolve or dissolve it. Keeping this line drawn is not caution for its own sake; it is the condition under which the measured core carries weight. A skeptic who is told “the conductor is the $\sqrt{N}$ read-budget, measured across two L-function families” can check it; a skeptic who is told “the universe is an FPGA and the halting problem is dissolved” has been given nothing to check. The program’s strength is the checkable part, and the interpretive frame is scaffolding around it.

9. “Change and go nowhere” — the measured invariant

One phrase from the discussion does name a measured structure precisely, and it belongs in the record on the measured side. “Change and go nowhere” is invariance under a transformation that moves the representation while fixing the object, and the program has it three ways. Under the isospectral (Toda) flow the recurrence coefficients move violently — the diagonal migrates, the off-diagonal collapses — while the wall and the count sit at the same index: the representation flows, the object goes nowhere. Under the affine read-group the coordinate frame translates, dilates, and reflects while the count is exact under all of it. And below the precision floor the machine burns digits computing sign after sign while the object’s value does not move — change paid for in exhaust, the state fixed. The rank read as the quadrature weight parked at the spectral origin is the same shape: the fixed point that absorbs the constraint while the non-central structure spreads and decays. These are theorems and measurements, and the phrase compresses them well. It is the invariant/execution split of the whole program, and it is exactly the reason the object is coordinate-free while its cost is not.

10. The frontier

Two targets remain, and both are now sharply stated rather than vaguely gestured at. On the BSD side, the open problem is the coefficient *certificate* under the instrument’s budget — the E-P6 window — which Section 4 locates as the question of how the Stieltjes instrument accommodates a $\sqrt{N}$ -length read when the conductor is large. It is a budget-versus-window problem, and the $\sqrt{N}$ law is the budget side of it made explicit. On the RH side, the target is the two-reading split on the prime-Stieltjes measure for zeta — a shape reading (the wall as a self-measuring counter) and an arithmetic-weighted reading (the wall tracking the zero-counting function $N(T)$) — the same structure the BSD channels realize, now on zeta. Section 5 has

already shown that zeta's read obeys the same $\sqrt{\text{conductor}}$ budget as the curve's, which is the first piece of that bridge; the remaining gate is the alignment of the moment convention between instances, so that the two readings are demonstrably reading the same object. Neither target is claimed here. Both are located.

11. Status ledger

claim	status	evidence
Weil bound = local RH, all five curves incl. rank 4	LOCKED (classical)	zero violations, $\max x_p \rightarrow 0.997$
order channel reads rank 0-4, monotone	PULL	Nagao ladder -0.22, 0.65, 1.83, 2.92, 3.34
coefficient channel reads $L^{(r)}/r!$ to rank 4	PULL	LMFDB to 10^{-10} (r 0-3), 5.3×10^{-6} (r 4)
fast G_r (incomplete-gamma) = slow quad	LOCKED (verified)	match to 10^{-31}
read-budget = $O(\sqrt{N})$ for elliptic curves	PULL	terms/ $\sqrt{N}$ flat ≈ 6 ; $\alpha = 0.5046$
read-budget = $\sqrt{t/2\pi}$ for zeta	LOCKED (classical) / PULL	Riemann-Siegel; error $\sim t^{-1/4}$; $\alpha = 0.536 \rightarrow 0.5$
$\sqrt{\text{conductor}}$ budget universal across families	LOCKED (analytic-conductor principle)	measured on two independent families
conductor lives in the budget, not the exhaust	PULL	budget object-determined; sub-floor object-blind (prior)
E-P6 gate = $\sqrt{N}$ budget vs compact-support window	PULL	rank-4 read-length ≈ 2000 + terms
"change and go nowhere" = invariance under representation flow	LOCKED / PULL	Toda, affine, sub-floor
instruction-set architecture = the measurements	Ω (interpretation)	three-layer sort is measured; ISA-as-ontology is not
halting problem dissolved	Ω (not established)	technical undecidability untouched by the reading

12. Reproduction

Every number was regenerated in-session. The accompanying scripts and notebooks reproduce each result: the two-channel BSD read to rank four, the $\sqrt{N}$ read-budget fit across five curves, the incomplete-gamma G_r verification against numerical integration, and the Riemann-Siegel main-sum budget for zeta across a range of heights, all under mpmath with each PASS/FAIL preregistered in the text that precedes it. Per the

program's standing discipline, no PULL row is promoted to LOCKED on the strength of this document; promotion waits on independent re-run, on separate hardware, preferably by a separate model instance.

Provenance. Produced within the multi-AI cross-validation network. The opcode/assembly framing, the cache-size reading of the conductor, and the instruction-set vocabulary entered through cross-model dialogue; the experiment designs, the runs, the $\sqrt{\text{conductor}}$ cross-family measurement, and the layer sorting are the working node's; direction, standards, and the anchor discipline are the author's. The contribution of the paper is a location and a law, not a claim on the open problems: the conductor is the $\sqrt{N}$ read-budget, the budget is universal across L-function families, and the friction the discussion sought is fetch length rather than exhaust noise.